

Multiple Choice (10 marks)

1. In Yann LeCun's cake, the cherry on top is
 - (a) reinforcement learning
 - (b) self-supervised learning
 - (c) unsupervised learning
 - (d) supervised learning
2. Statement 1| Highway networks were introduced after ResNets and eschew max pooling in favor of convolutions. Statement 2| DenseNets usually cost more memory than ResNets.
 - (a) True, True
 - (b) False, False
 - (c) True, False
 - (d) False, True
3. Which one of the following is the main reason for pruning a Decision Tree?
 - (a) To save computing time during testing
 - (b) To save space for storing the Decision Tree
 - (c) To make the training set error smaller
 - (d) To avoid overfitting the training set
4. Statement 1| The Stanford Sentiment Treebank contained movie reviews, not book reviews. Statement 2| The Penn Treebank has been used for language modeling.
 - (a) True, True
 - (b) False, False
 - (c) True, False
 - (d) False, True
5. A and B are two events. If $P(A, B)$ decreases while $P(A)$ increases, which of the following is true?
 - (a) $P(A|B)$ decreases
 - (b) $P(B|A)$ decreases
 - (c) $P(B)$ decreases
 - (d) All of above

True / False (10 marks)

Answer True or False

6. True or False: Transforming data to zero median is sufficient to achieve the same projection as SVD in PCA.
7. True or False: By₂₀₂₀, several models achieved accuracy levels exceeding 98% on the CIFAR-10 dataset.
8. True or False: Grid search runs reasonably slow for multiple linear regression due to its exhaustive nature in parameter tuning.
9. True or False: Convolutional networks are the most effective architecture for classifying high-resolution images as of 2020.
10. True or False: The Bayesian network $H \rightarrow U \leftarrow P \leftarrow W$ requires 8 independent parameters to fully characterize its structure.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Write a function to find the cumulative sum of all the values that are present in the given tuple list.
12. Write a python function to toggle all even bits of a given number.

End of Paper